

LSTM vs Transformer

Stock Price Prediction — Fair Comparison Report

Fresh inference run: 2026-04-26 01:34

Symbols evaluated: 17 / 17 • Test split: last 15% (chronological)

Avg MAPE → LSTM: 8.77% | Transformer: 8.72%

Avg R² → LSTM: -2.408 | Transformer: -2.126

Avg DirAcc → LSTM: 49.4% | Transformer: 49.3%

Architecture — LSTM:

3 × LSTM layers (128 → 64 → 32 units) + BatchNorm + Dropout + Dense head

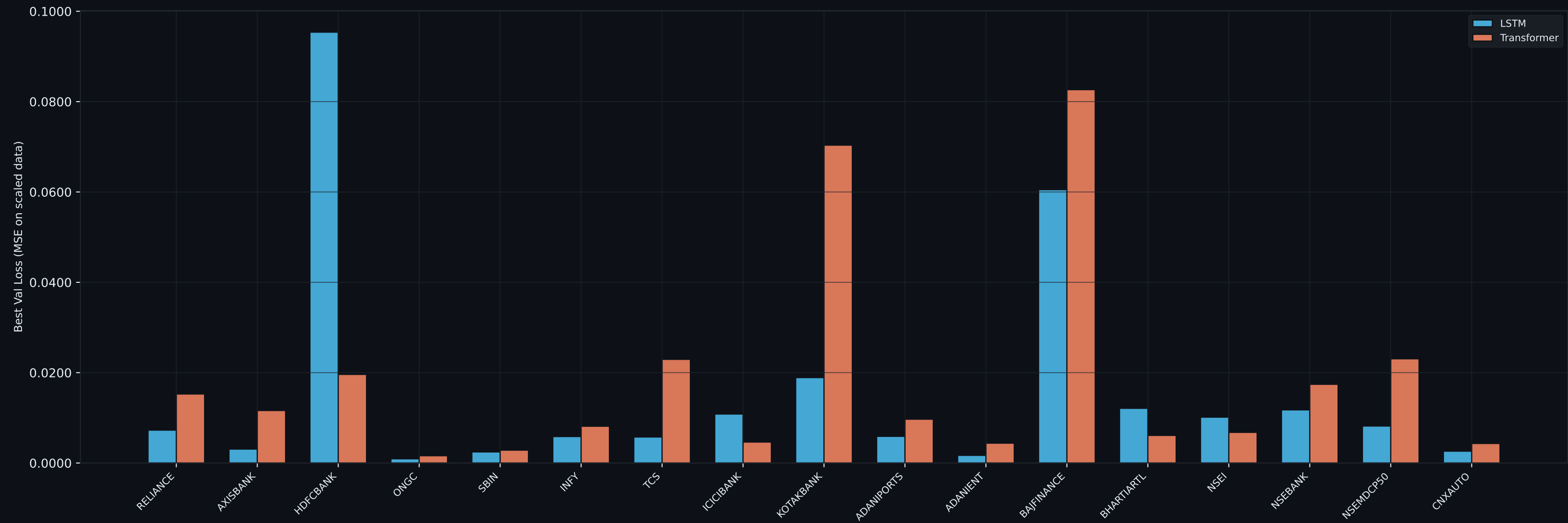
Architecture — Transformer:

3 × Encoder blocks (Multi-Head Attention 8h×64 + GELU FFN 256) + Positional Embedding

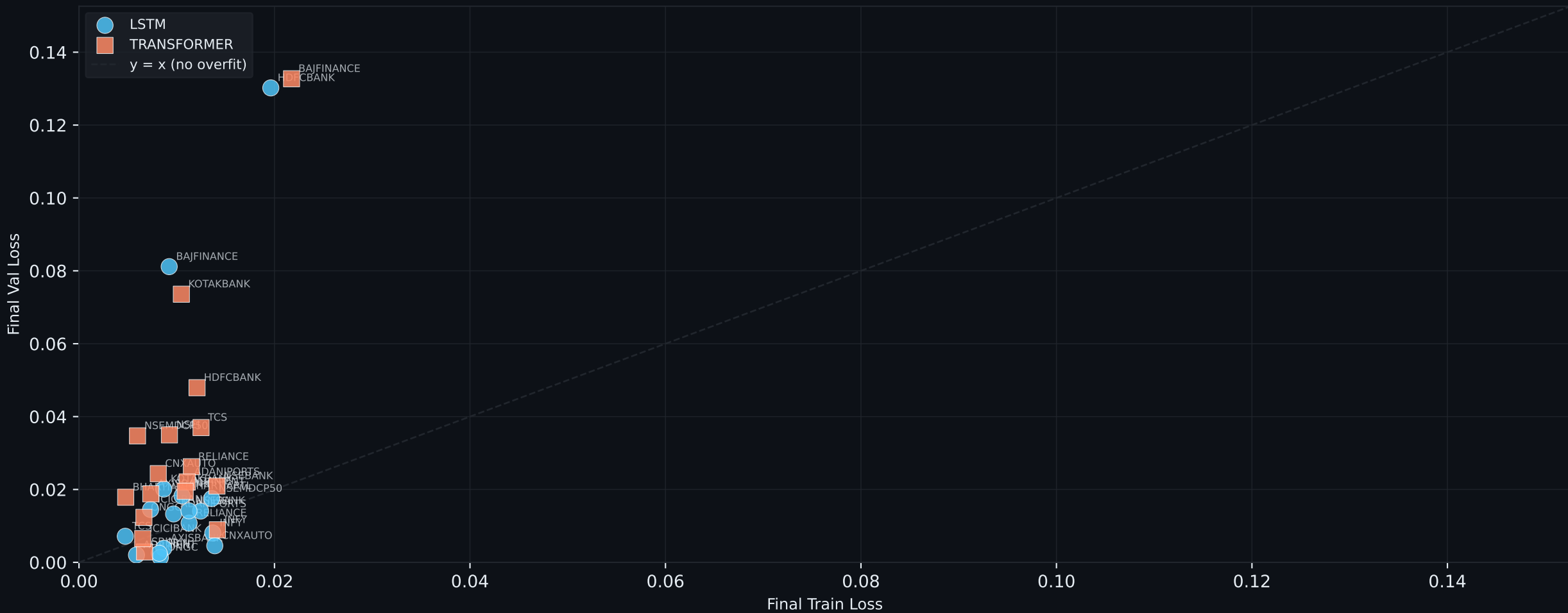
+ GlobalAveragePooling + MLP head (128 → 64 → 1)

Train/Val/Test: 70%/15%/15% chronological • Seq len: 120 • Seed: 42 • Period: 5 years

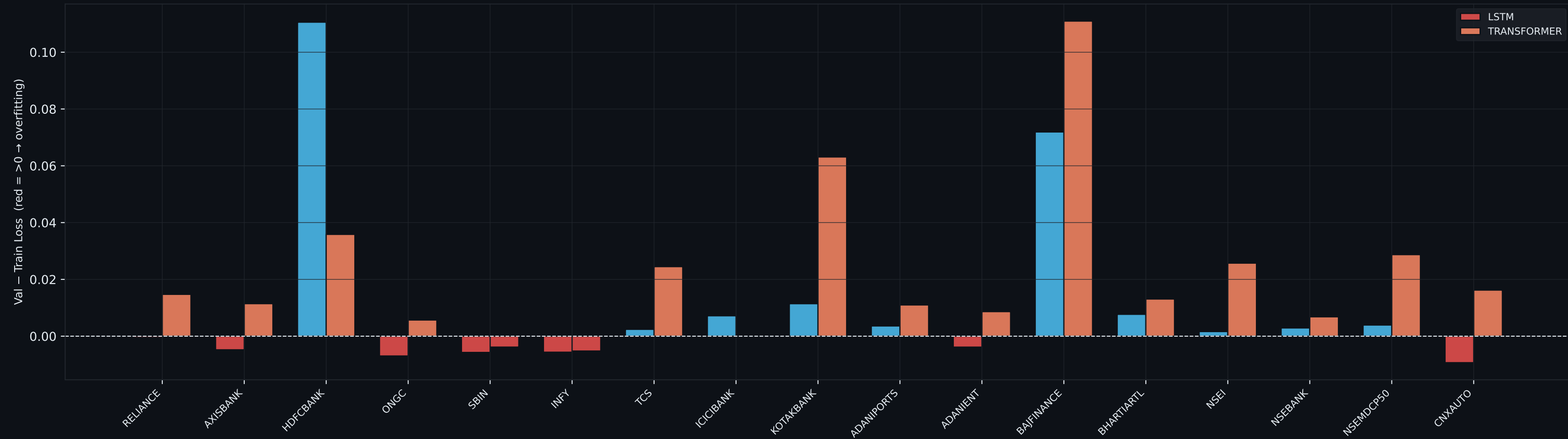
Training Summary — Best Validation Loss per Symbol



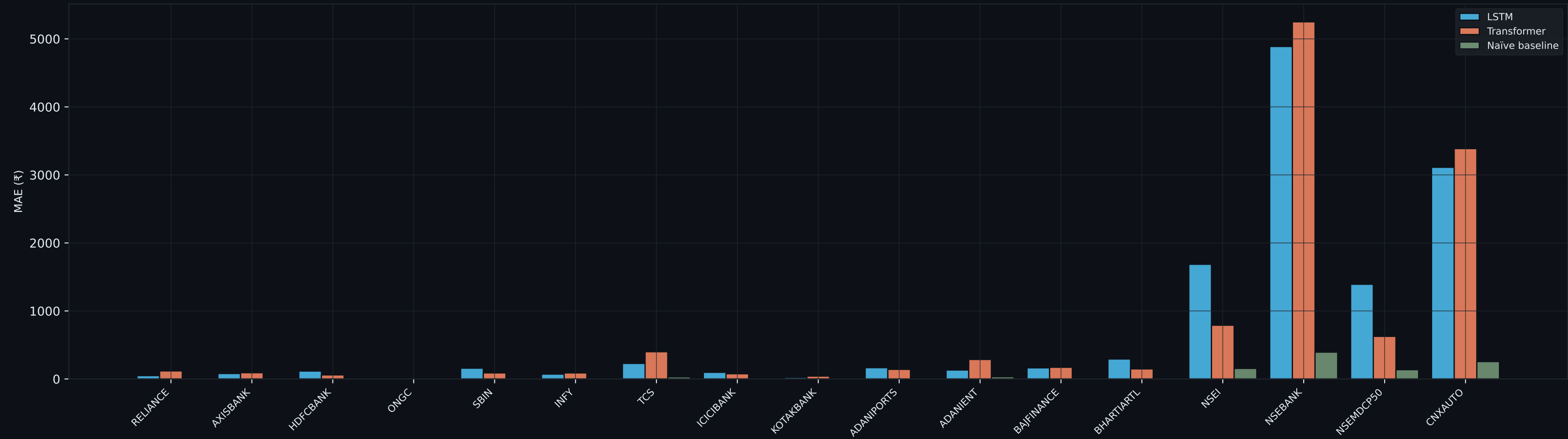
Overfitting Check — Final Train Loss vs Final Val Loss



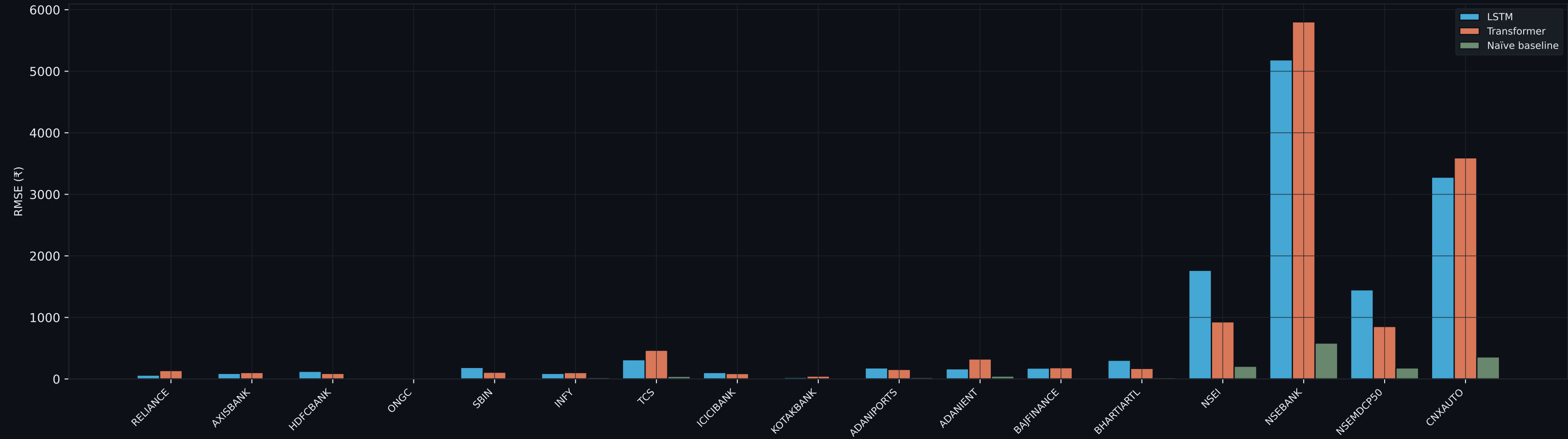
Train-Val Loss Gap (Val – Train) — Overfitting Indicator



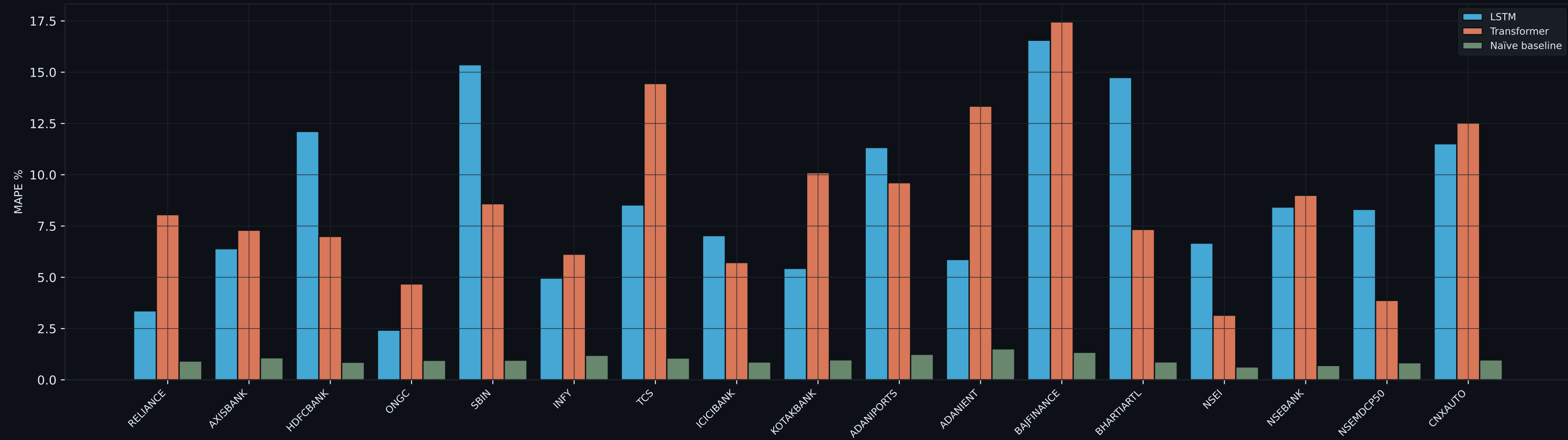
Model Comparison — MAE (₹)



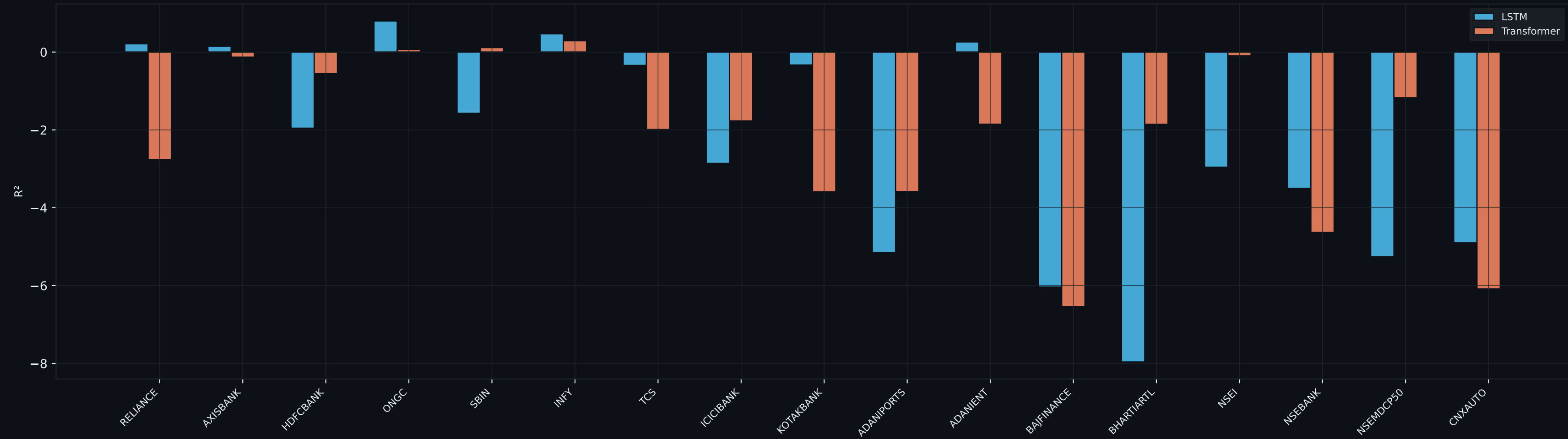
Model Comparison — RMSE (₹)



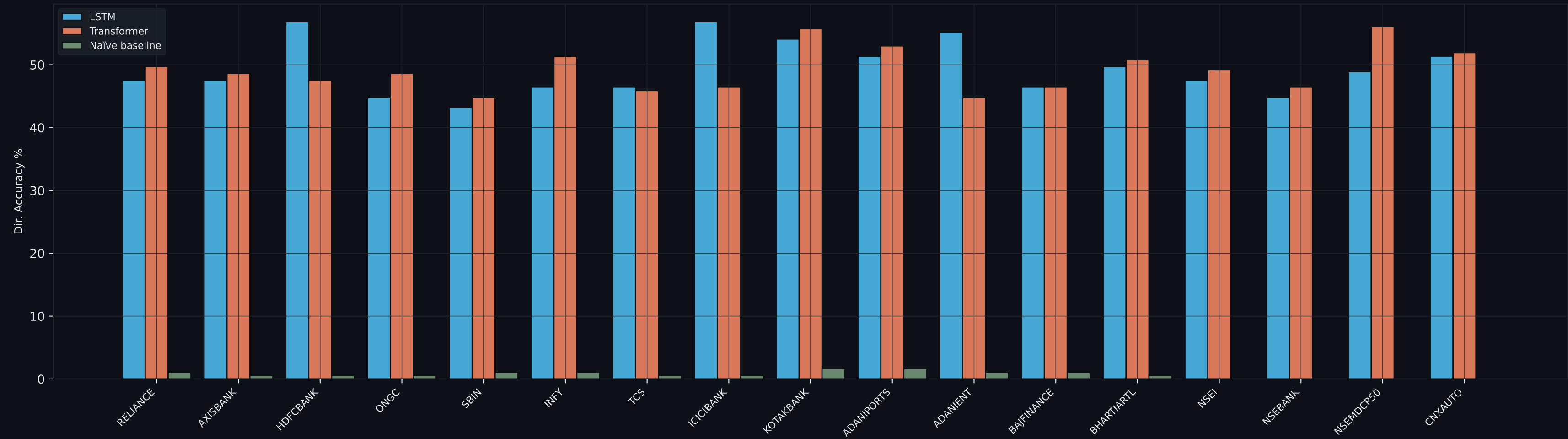
Model Comparison — MAPE %



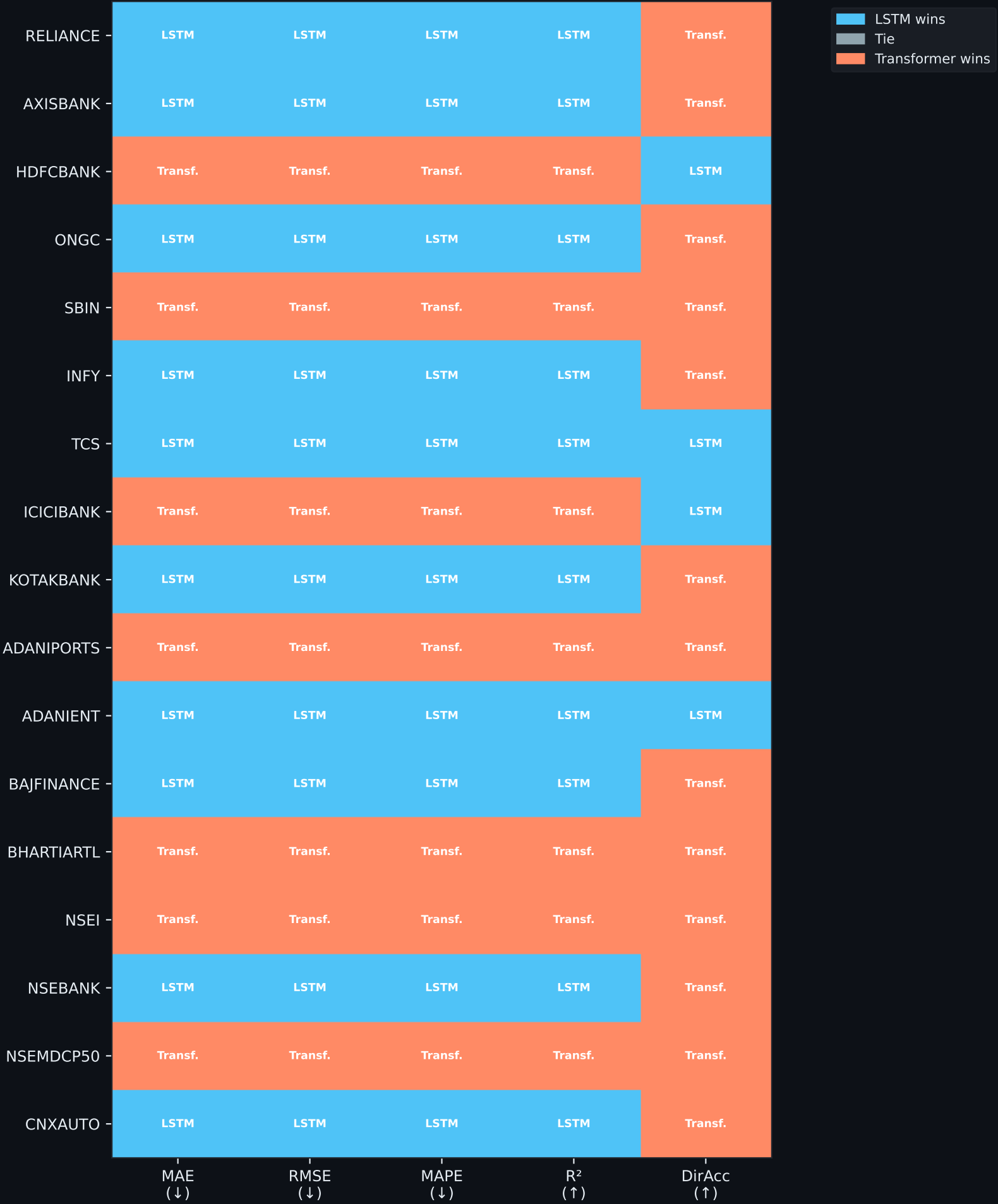
Model Comparison — R²



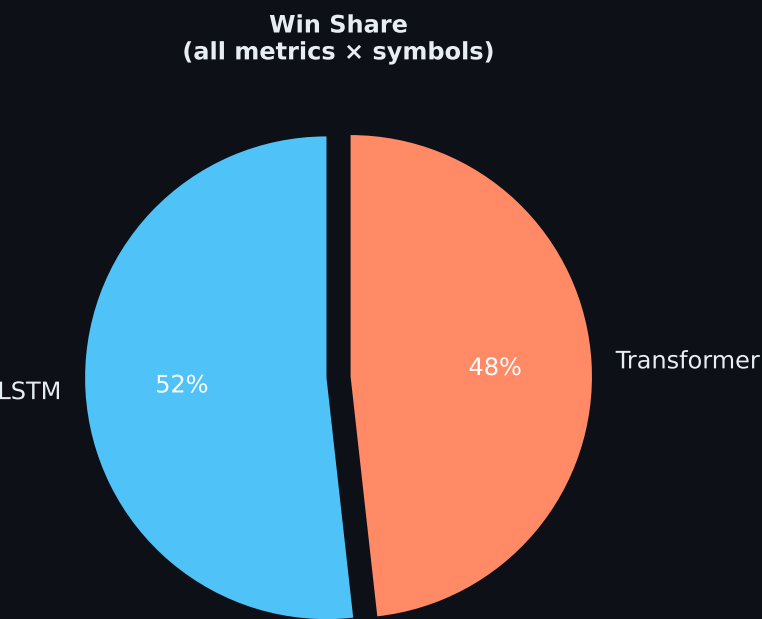
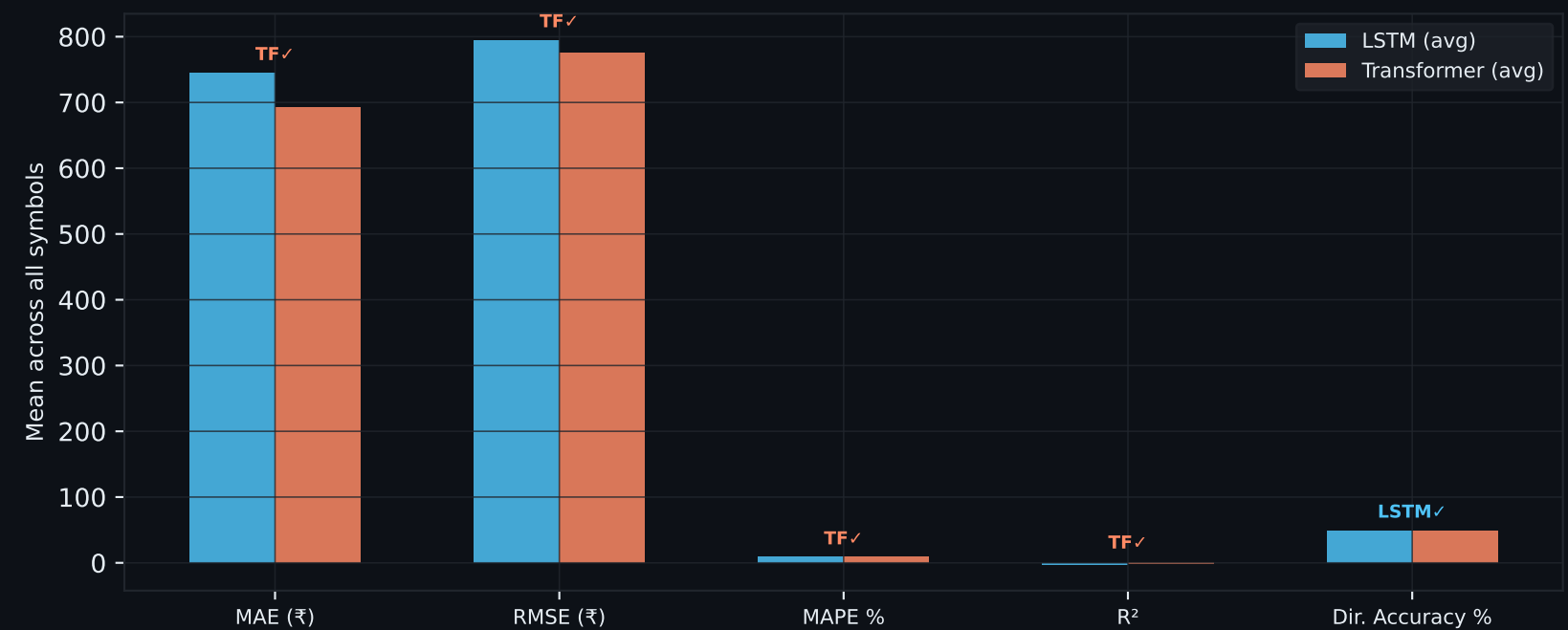
Model Comparison — Dir. Accuracy %



Win Heatmap — LSTM vs Transformer



Aggregate Performance Summary

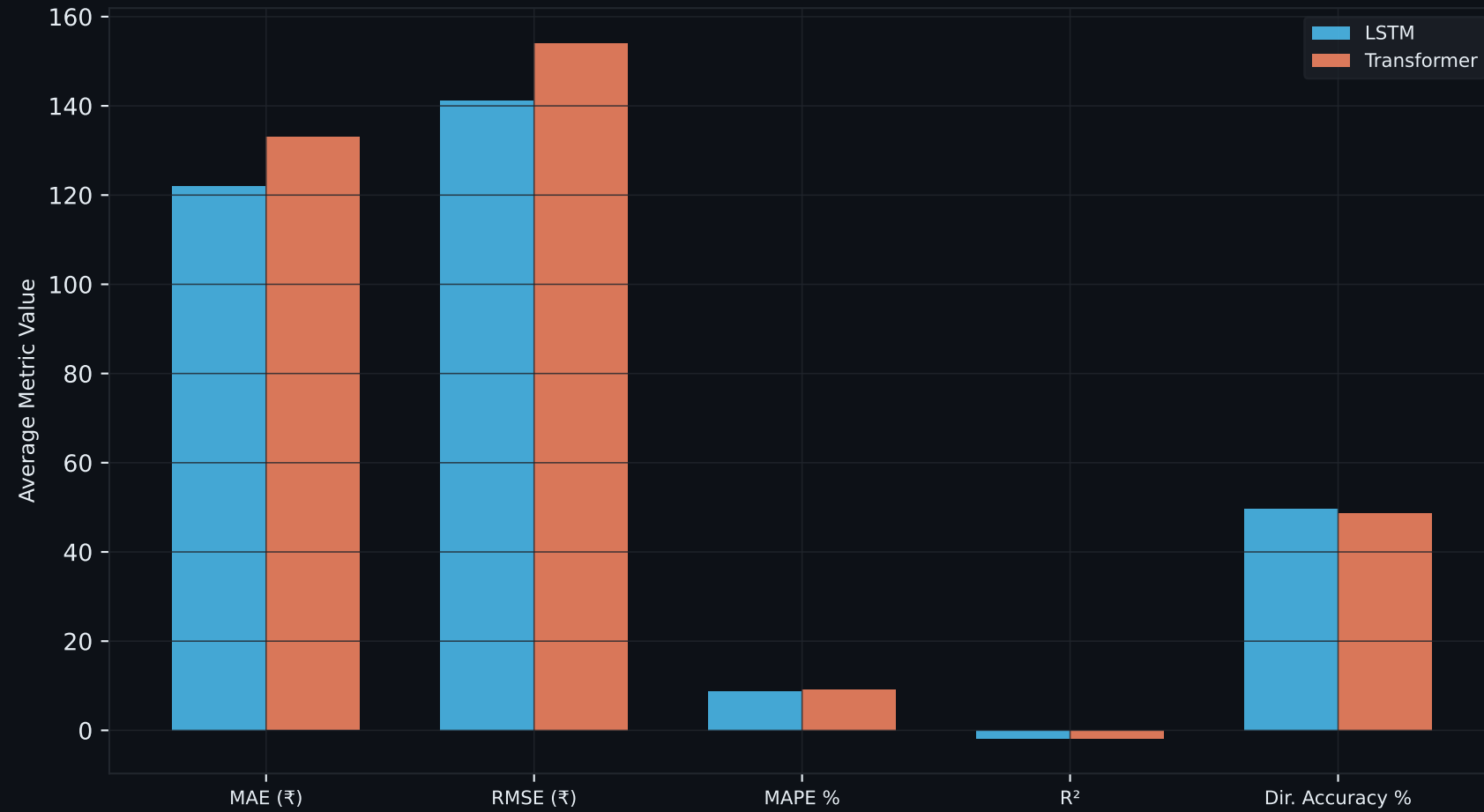


MAPE % Distribution (Test Split)

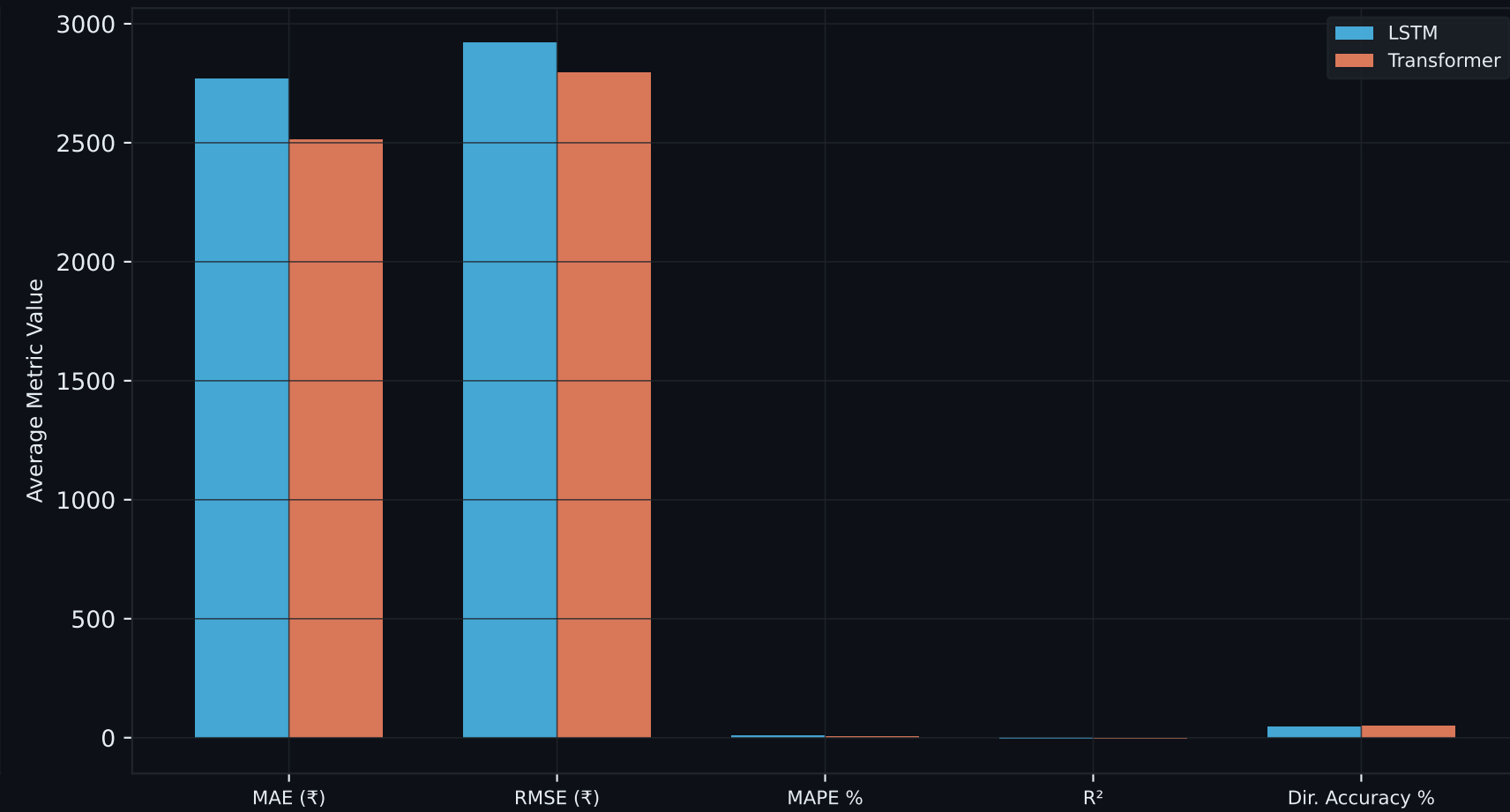


Average Metrics — Stocks vs Indices

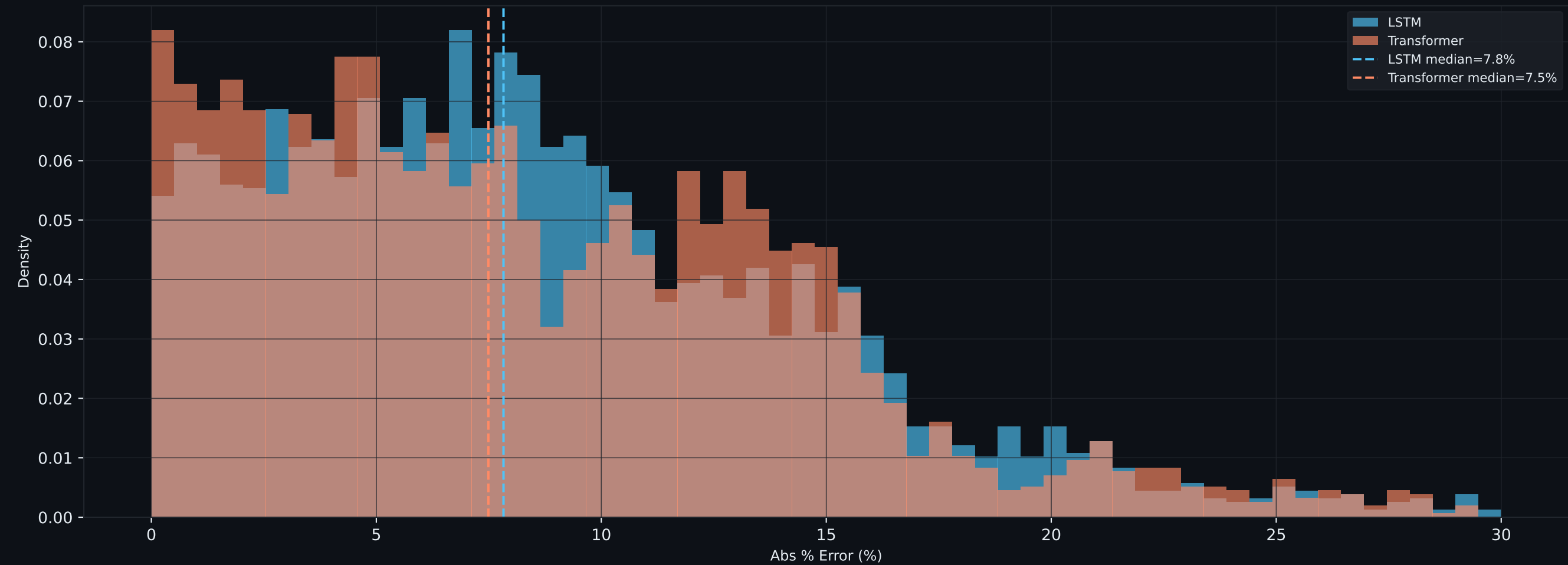
Stocks



Indices



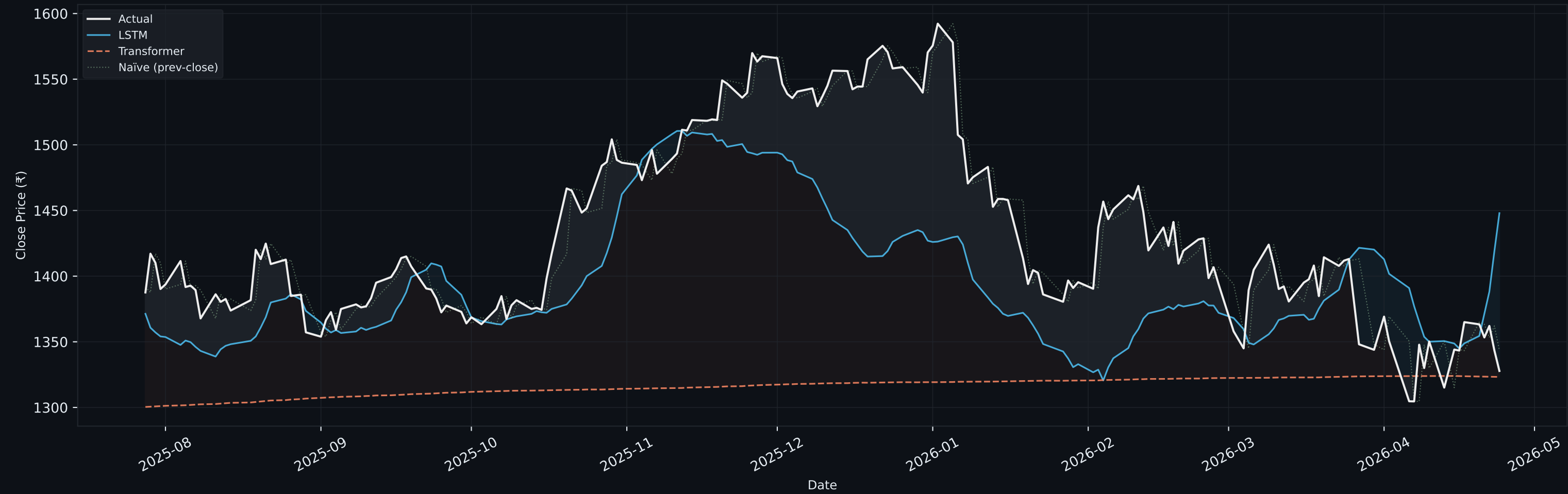
Absolute % Error Distribution Across All Symbols (Test Split)



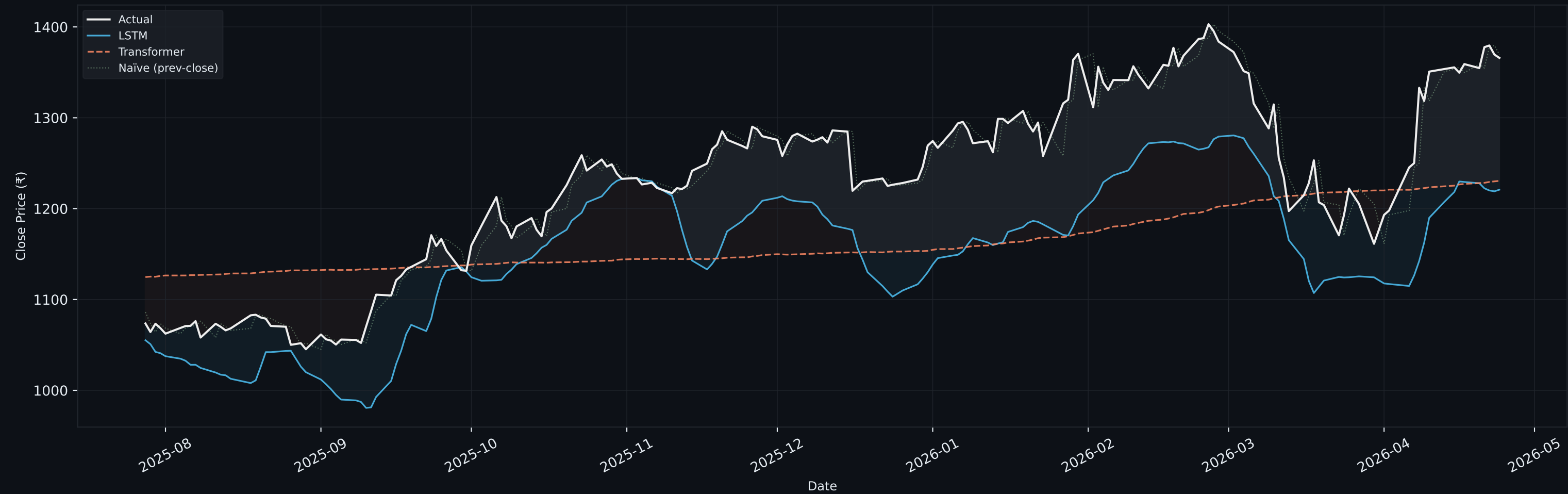
Full Metrics Table — Fresh Inference (Test Split)

Symbol	MAE LSTM	MAE Transf.	RMSE LSTM	RMSE Transf.	MAPE LSTM	MAPE Transf.	R² LSTM	R² Transf.	DirAcc LSTM	DirAcc Transf.	Naive MAE
RELIANCE	49.2	118.5	62.9	137.4	3.36%	8.05%	0.212	-2.761	47.5%	49.7%	13.0
AXISBANK	80.2	92.1	90.9	105.0	6.39%	7.30%	0.152	-0.131	47.5%	48.6%	13.3
HDFCBANK	116.5	59.6	124.8	90.6	12.11%	6.99%	-1.956	-0.560	56.8%	47.5%	7.7
ONGC	6.2	12.4	7.8	16.7	2.42%	4.68%	0.797	0.069	44.8%	48.6%	2.4
SBIN	158.9	89.1	187.8	110.1	15.37%	8.59%	-1.573	0.115	43.2%	44.8%	9.6
INFY	71.6	89.0	90.6	104.7	4.96%	6.13%	0.468	0.291	46.4%	51.4%	17.4
TCS	228.7	400.5	313.0	466.9	8.53%	14.45%	-0.344	-1.989	46.4%	45.9%	30.8
ICICIBANK	97.8	76.8	105.1	89.0	7.03%	5.72%	-2.861	-1.771	56.8%	46.4%	11.8
KOTAKBANK	22.7	42.3	25.4	47.1	5.44%	10.10%	-0.333	-3.590	54.1%	55.7%	3.9
ADANIPTS	166.1	141.4	180.2	155.6	11.33%	9.61%	-5.151	-3.583	51.4%	53.0%	17.8
ADANIPT	131.9	286.8	165.5	324.9	5.87%	13.34%	0.259	-1.854	55.2%	44.8%	33.4
BAJFINANCE	163.7	172.0	177.4	183.6	16.56%	17.45%	-6.035	-6.535	46.4%	46.4%	12.7
BHARTIARTL	293.6	148.3	304.6	172.0	14.74%	7.34%	-7.961	-1.857	49.7%	50.8%	17.2
NSEI	1686.3	790.5	1765.7	928.4	6.67%	3.15%	-2.958	-0.094	47.5%	49.2%	155.2
NSEBANK	4888.1	5250.2	5184.9	5802.6	8.42%	9.00%	-3.500	-4.636	44.8%	46.4%	394.5
NSEMDCP50	1392.6	626.4	1448.6	853.7	8.31%	3.87%	-5.257	-1.173	48.9%	56.0%	136.7
CNXAUTO	3112.0	3386.9	3279.4	3593.3	11.51%	12.54%	-4.901	-6.085	51.4%	51.9%	255.9

Test-Split Predictions — RELIANCE.NS



Test-Split Predictions — AXISBANK.NS



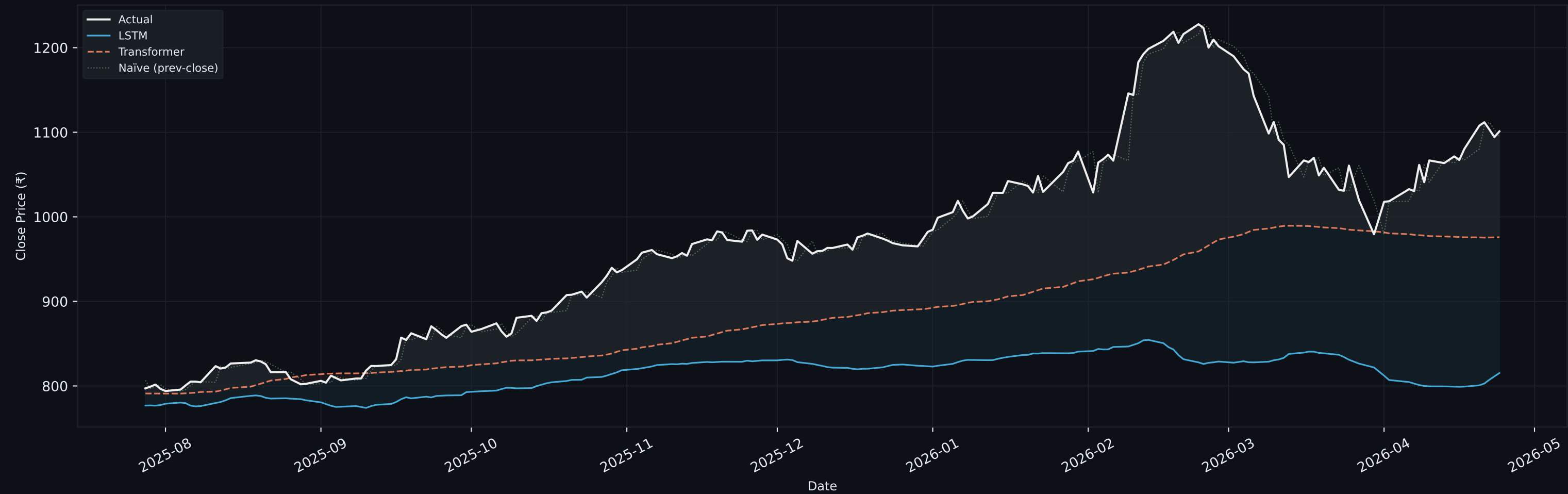
Test-Split Predictions — HDFCBANK.NS



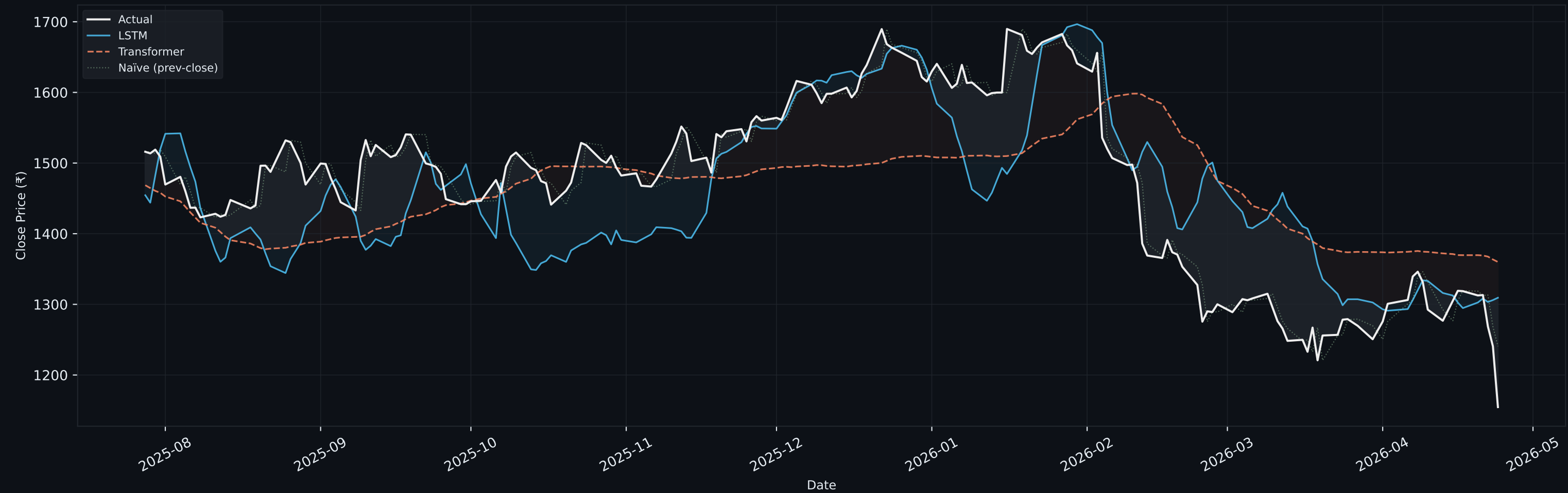
Test-Split Predictions — ONGC.NS



Test-Split Predictions — SBIN.NS



Test-Split Predictions — INFY.NS



Test-Split Predictions — TCS.NS



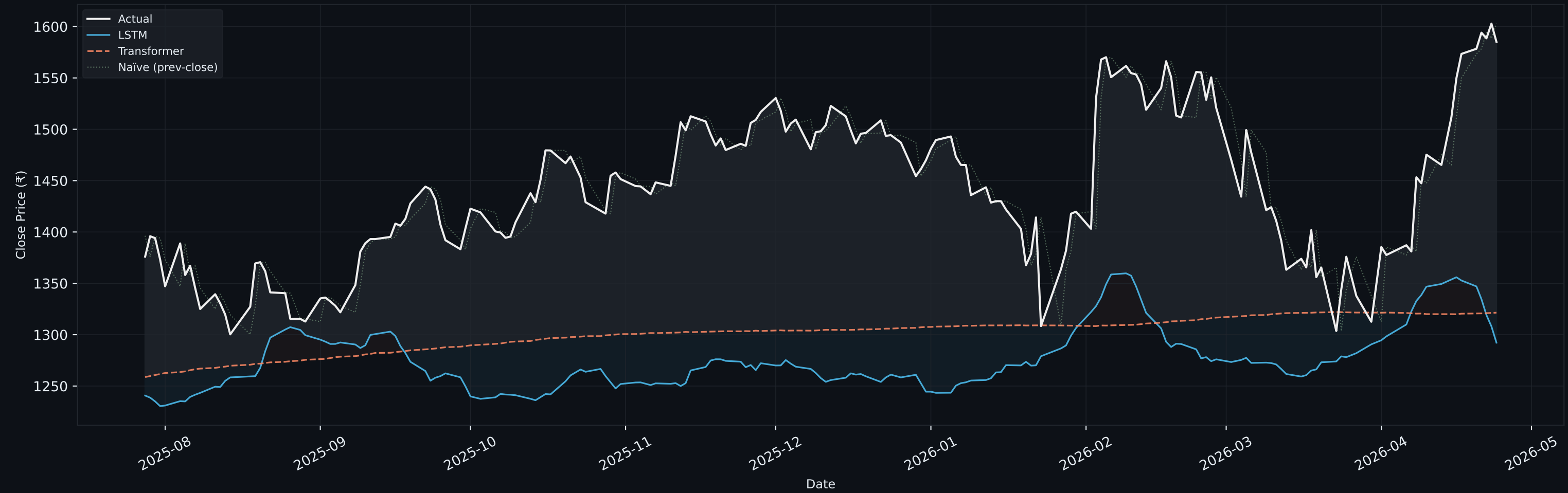
Test-Split Predictions — ICICIBANK.NS



Test-Split Predictions — KOTAKBANK.NS



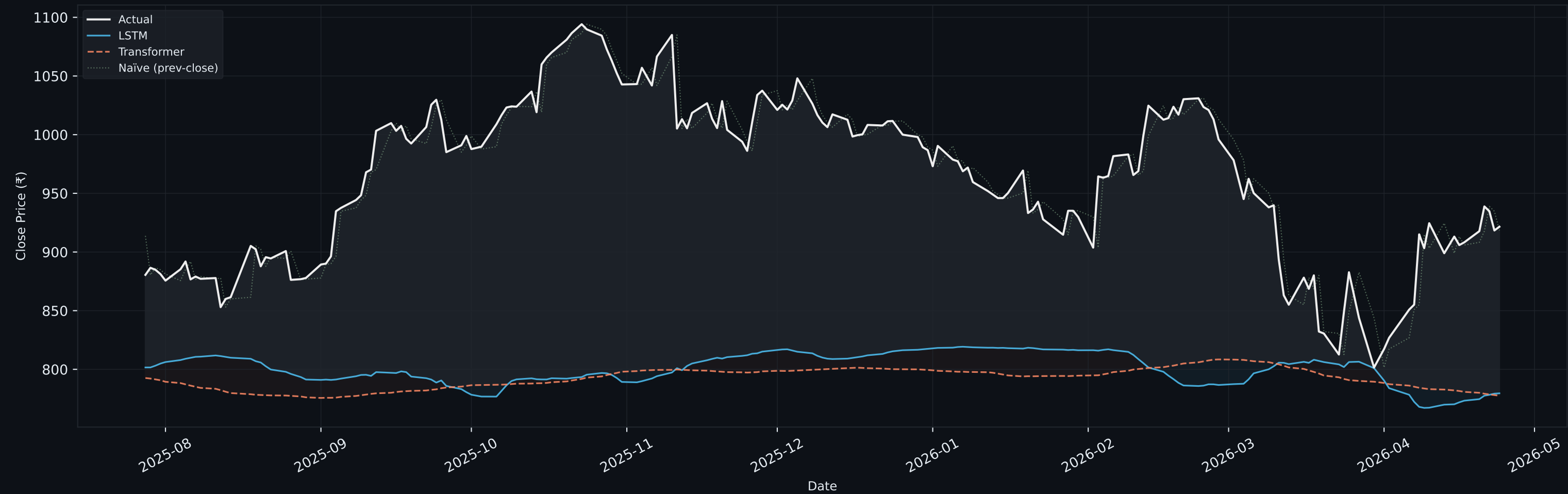
Test-Split Predictions — ADANI PORTS.NS



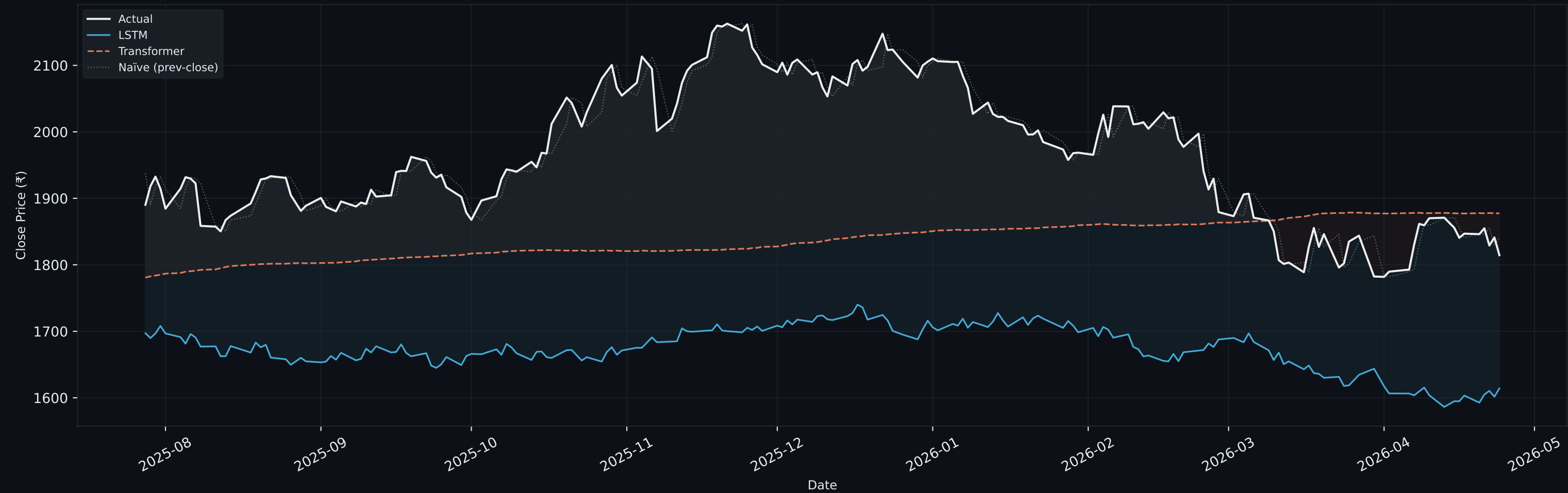
Test-Split Predictions — ADANIENT.NS



Test-Split Predictions — BAJFINANCE.NS



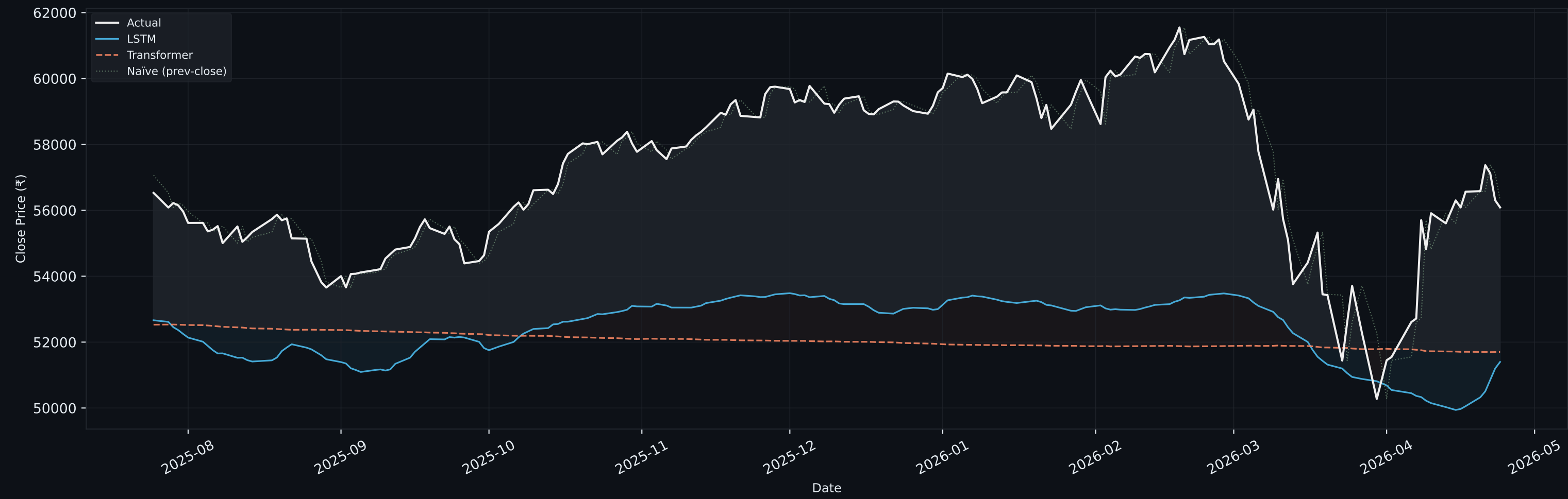
Test-Split Predictions — BHARTIARTL.NS



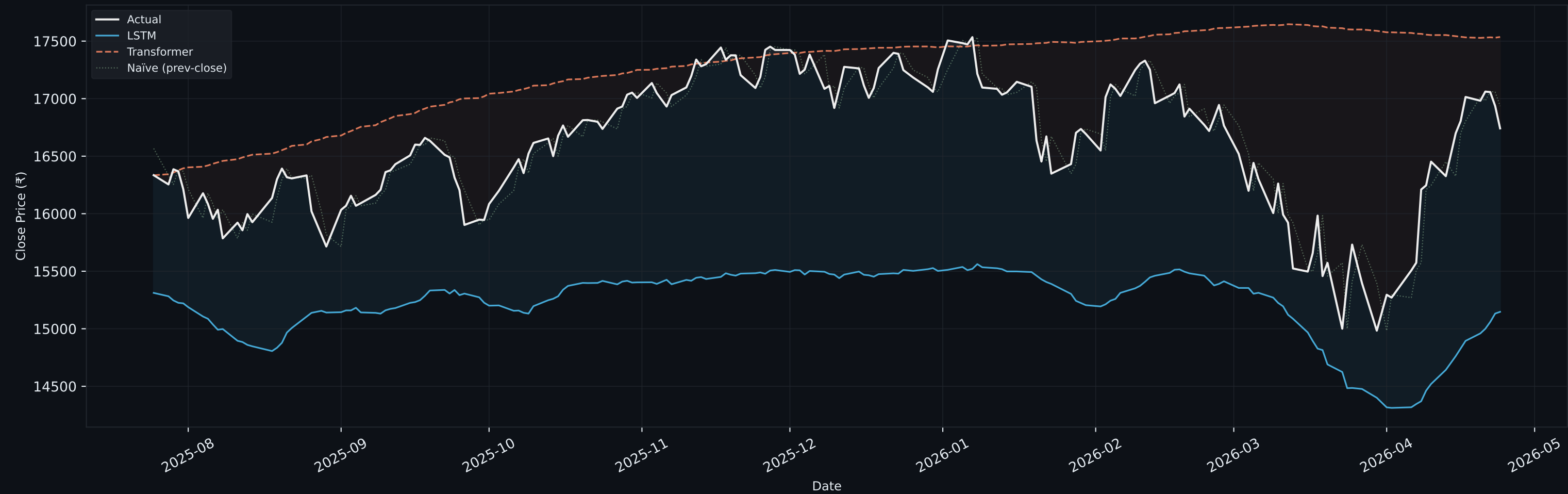
Test-Split Predictions — ^NSEI



Test-Split Predictions — ^NSEBANK



Test-Split Predictions — $\hat{NSEMDCP50}$



Test-Split Predictions — ^CNXAUTO

